MarinEARS – Marine Explorer and Registry of Sound

Datenanforderungen für Seismische Untersuchungen

Bundesamt für Seeschifffahrt und Hydrographie

Hamburg und Rostock

® Bundesamt für Seeschifffahrt und Hydrographie (BSH)
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www.bsh.de

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Einleitung

Daten und Informationen über Impulsschallereignisse durch seismische Untersuchungen sind an das nationale Schallregister zu berichten, welches in das Hydroschall-Fachinformationssystem MarinEARS des BSH integriert ist. MarinEARS ermöglicht die elektronische Erfassung von prozessierten Messdaten, Metadaten und Unterlagen zu Messungen von Unterwasserschall. Derzeit erfolgt die Einreichung von Daten zu seismischen Untersuchungen per E-Mail an MarinEARS@bsh.de. Dateien > 10MB sind über Cloud-Dienste einzureichen.

Die Berichterstattung an das BSH bezüglich des durch seismische Untersuchungen verursachten Unterwasserschalls umfasst die in diesem Dokument genannten Parameter. Alle verwendeten Geräte müssen für die Schallbetrachtung berichtet werden.

Die Metadaten sind in einem HDF5 mit der im folgenden dargestellten Struktur einzureichen. In dem HDF5 werden in den Gruppen „airguns“, „boomer“, „sparker“ und „sub_bottom_profiler“ alle verwendeten Geräte aufgenommen. Falls sich die Geräteparameter im Laufe der Ausfahrt ändern, werden diese Geräte mehrfach aufgenommen und den Geräte-Labeln ein Buchstabe hinzugefügt. In der Gruppe „transects“ werden jedem Fahrtabschnitt die eingesetzten Geräte zugewiesen. Eine beispielhafte HDF5-Struktur befindet sich am Ende dieser Anleitung.
Zusätzlich ist für die gesamte Ausfahrt anzugeben, an welchen Tagen in welchen ICES-SubRectangles Schall welcher Lautstärkekategorie emittiert wurde. Das ICES nutzt hierfür folgende Klassifizierung, wobei Sparker, Boomer und Sub-bottom profiler in die linke Kategorie und Airguns in die rechte Kategorie fallen:

value_code	Generic explicitly impulsive source (energy source level, rounded to nearest decibel)	Airgun arrays (zero to peak source level, rounded to nearest decibel):
NA		
very_low	186-210 dB re 1 µPa2 m2 s	209-233 dB re 1 µPa m
low	211-220 dB re 1 µPa2 m2 s	234-243 dB re 1 µPa m
medium	221-230 dB re 1 µPa2 m2 s	244-253 dB re 1 µPa m
high	> 230 dB re 1 µPa2 m2 s	> 253 dB re 1 µPa m
very_high		

Datenübermittlungen zu Messungen des Unterwasserschalls sind separat darzustellen und in einem geeigneten Format (möglichst HDF5 oder Matlab) einzureichen.

HDF5-Struktur für Metadaten von seismische Untersuchungen

Field	Status	Data type	Field definition	Example
/				
author	Mandatory	string	Creator of the hdf5 file/ who holds responsibility for data QA and creation of the submitted hdf5 file	Vorname.Name@Domain.de
comment	Optional	string		
date_of_creation	Mandatory	string	Date of file creation, UTC DateTime in ISO 8601 format; YYYY-MM-DD	e.g: 2021-09-17
journey_name	Mandatory	string	Name of the journey	e.g.: Seismic_Survey_XYZ_2021
measurement_purpose	Mandatory	string	Description for what purpose the journey was made	
measuring_institution	Mandatory	string	Institution, which acquired the data	
point_of_contact	Mandatory	string	Contact for all future external queries/ who submits/holds responsibility for submission	Beispielinstitut@Domain.de
/airguns	Optional	group		
airgun_count	Mandatory	integer	Number of airguns and airgun configurations	e.g.: 3
/airgun_1	Optional	group	Different airgun configurations are indicated by adding a letter	e.g.:airgun_2a, airgun_2b
beam_width	Optional	float	half-power beam width [°]	
comment	Optional	string		
manufacturer	Mandatory	string	airgun manufacturer	
max_signal_length	Optional	float	Maximal recorded signal length including multiple reflections; time interval which contains 95% of the recorded energy. Use a nearby receiver with good data quality. Use a shot at a position of the maximal signal length of the transect. Create multiple airgun configuration entrys for the case that there are large variations in signal length (e.g. due to water depth variations or varying reflection coefficients).	e.g.: 3.1
name	Optional	string	airgun type	e.g.: 1500LL
frequency_min_theo	Optional	float	Minimum of the possible frequency band [Hz] at which the energy has dropped by 6 dB according to manufacturer	e.g.: 10.0
frequency_max_theo	Optional	float	Maximum of the possible frequency band [Hz] at which the energy has dropped by 6 dB according to manufacturer	e.g.: 120.0
frequency_min_observed	Mandatory	float	Minimum of measured frequency band at which the energy has dropped by 6 dB [Hz], if possible use the direct wave and a nearby receiver with good data quality.	e.g.: 12.0
frequency_max_observed	Mandatory	float	Maximum of measured frequency band [Hz] at which the energy has dropped by 6 dB, if possible use the direct wave and a nearby receiver with good data quality.	e.g.: 121.0
frequency_spectrum	Optional	float(number of entries x 2)	Frequency [Hz] and corresponding measured power spectral density, if possible use the direct wave and a nearby receiver with good data quality.	
pressure	Mandatory	float	Operating pressure [bar]	
pulse_rate	Mandatory	float	pulse rate [Hz]	e.g.: 0.25
pulse_length	Mandatory	float	Time length of source pulse [s], where 95% of the energy is included	e.g.: 0.01
serial_number	Optional	string	Serial number of airgun	
source_depth	Mandatory	float	Depth of airgun below the water surface [m]	e.g.: 1.5
source_level_Lpeak	Optional	integer	Zero-to-peak level [dB re 1 µPa] at 1m distance	e.g.: 220
source_level_SEL	Optional	integer	Sound-exposure level [dB re 1 µPa^2s] at 1m distance	e.g.: 190
volume_generator	Mandatory	float	Air chamber volume of generator [Liter]	
volume_injector	Optional	float	Air chamber volume of injector [Liter]	
/boomer	Optional	group		
boomer_count	Mandatory	integer	Number of boomers and boomer configurations	e.g.: 3
/boomer_1	Optional	group	Different boomer configurations are indicated by adding a letter	e.g.: boomer_1a, boomer_1b
beam_width	Optional	float	half-power beam width phi [°]	
comment	Optional	string		
manufacturer	Mandatory	string	boomer manufacturer	
max_signal_length	Optional	float	Maximal recorded signal length including multiple reflections. Compute the time which contains 95% of the energy. Use a nearby receiver with good data quality. Use a shot at a position of the maximal signal length of the transect. Create multiple boomer configuration entrys for the case that there are large variations in signal length (e.g. due to water depth variations or varying reflection coefficients).	e.g.: 3.1
name	Optional	string	boomer type	e.g.: AA251
electric_energy_max	Mandatory	float	Maximum electric energy [J]	
electric_power_max	Optional	float	Maximum electric power [kW]	
frequency_min_theo	Optional	float	Minimum of the possible frequency band [Hz] at which the energy has dropped by 6 dB according to manufacturer	e.g.: 10.0
frequency_max_theo	Optional	float	Maximum of the possible frequency band [Hz] at which the energy has dropped by 6 dB according to manufacturer	e.g.: 120.0
frequency_min_observed	Mandatory	float	Minimum of measured frequency band at which the energy has dropped by 6 dB [Hz], if possible use the direct wave and a nearby receiver with good data quality.	e.g.: 12.0
frequency_max_observed	Mandatory	float	Maximum of measured frequency band [Hz] at which the energy has dropped by 6 dB, if possible use the direct wave and a nearby receiver with good data quality.	e.g.: 121.0

frequency_spectrum	Optional	float(number of entries x 2)	Frequency [Hz] and corresponding measured power spectral density, if possible use the direct wave and a nearby receiver with good data quality.	
plate_area	Mandatory	float	Area of the boomer plate [m^2]	e.g.: 0.1
pulse_rate	Mandatory	float	pulse rate [Hz]	e.g.: 0.25
pulse_length	Mandatory	float	Time length of source pulse [s], where 95% of the energy is included	e.g.: 0.01
serial_number	Optional	string	Serial number of boomer	
source_depth	Mandatory	float	Depth of boomer plate below the water surface [m]	e.g.: 1.5
source_level_Lpeak	Optional	integer	Zero-to-peak level [dB re 1 µPa] at 1m distance	e.g.: 212
source_level_SEL	Optional	integer	Sound-exposure level [dB re 1 µPa^2s] at 1m distance	e.g.: 190
/sparker	Optional	group		
sparker_count	Mandatory	integer	Number of sparkers and sparker configurations	e.g.: 3
/sparker_1	Optional	group	Different sparker configurations are indicated by adding a letter	e.g.: sparker_3a, sparker_3b
beam_width	Optional	float	half-power beam width phi [°]	
comment	Optional	string		
manufacturer	Mandatory	string	sparker manufacturer	
max_signal_length	Optional	float	Maximal recorded signal length including multiple reflections. Compute the time which contains 95% of the energy. Use a nearby receiver with good data quality. Use a shot at a position of the maximal signal length of the transect. Create multiple sparker configuration entrys for the case that there are large variations in signal length (e.g. due to water depth variations or varying reflection coefficients).	e.g.: 3.1
name	Optional	string	sparker type	e.g.: Dura-Spark L80
electric_energy_max	Mandatory	float	Maximum electric energy [J]	
electric_power_max	Optional	float	Maximum electric power [kW]	
frequency_min_theo	Optional	float	Minimum of the possible frequency band [Hz] at which the energy has dropped by 6 dB according to manufacturer	e.g.: 10.0
frequency_max_theo	Optional	float	Maximum of the possible frequency band [Hz] at which the energy has dropped by 6 dB according to manufacturer	e.g.: 120.0
frequency_min_observed	Mandatory	float	Minimum of measured frequency band at which the energy has dropped by 6 dB [Hz], if possible use the direct wave and a nearby receiver with good data quality.	e.g.: 12.0
frequency_max_observed	Mandatory	float	Maximum of measured frequency band [Hz] at which the energy has dropped by 6 dB, if possible use the direct wave and a nearby receiver with good data quality.	e.g.: 121
frequency_spectrum	Optional	float(number of entries x 2)	Frequency [Hz] and corresponding measured power spectral density, if possible use the direct wave and a nearby receiver with good data quality.	
pulse_rate	Mandatory	float	pulse rate [Hz]	e.g.: 0.25
pulse_length	Mandatory	float	Time length of source pulse [s], where 95% of the energy is included	e.g.: 0.01
serial_number	Optional	string	Serial number of sparker	
source_depth	Mandatory	float	Depth of sparker below the water surface [m]	e.g.: 1.5
source_level_Lpeak	Optional	integer	Zero-to-peak level [dB re 1 µPa] at 1m distance	e.g.: 220
source_level_SEL	Optional	integer	Sound-exposure level [dB re 1 µPa^2s] at 1m distance	e.g.: 195
/sub_bottom_profiler	Optional	group		
sbp_count	Mandatory	integer	Number of sub-bottom profilers and sub-bottom profiler configurations	e.g.: 3
/sbp_1	Optional	group	Different sub-bottom profiler configurations are indicated by adding a letter	e.g.: sbp_3a, sbp_3b
beam_width	Mandatory	float	half-power beam width phi [°]	e.g.: 3.0
comment	Optional	string		
manufacturer	Mandatory	string		
max_signal_length	Optional	float	Maximal recorded signal length including multiple reflections. Compute the time which contains 95% of the energy. Use a nearby receiver with good data quality, where the direct wave and reflected wave are already sepearated. Use a shot at a position of the maximal signal length of the transect. Create multiple sub-bottom profiler configuration entrys for the case that there are large variations in signal length (e.g. due to water depth variations or varying reflection coefficients).	e.g.: 3.1
name	Optional	string	sub-bottom profiler type	e.g.: 2000-DSS
electric_energy_max	Optional	float	Maximum electric energy [J]	
electric_power_max	Optional	float	Maximum electric power [kW]	
frequency_min_theo	Optional	float	Minimum of the possible frequency band [Hz] at which the energy has dropped by 6 dB according to manufacturer	e.g.: 10.0
frequency_max_theo	Optional	float	Maximum of the possible frequency band [Hz] at which the energy has dropped by 6 dB according to manufacturer	e.g.: 120.0
frequency_min_observed	Mandatory	float	Minimum of measured frequency band at which the energy has dropped by 6 dB [Hz]	e.g.: 12.0
frequency_max_observed	Mandatory	float	Maximum of measured frequency band [Hz] at which the energy has dropped by 6 dB	e.g.: 121.0
frequency_spectrum	Optional	float(number of entries x 2)	Frequency [Hz] and corresponding measured power spectral density	
pulse_rate	Mandatory	float	pulse rate [Hz]	e.g.: 0.25
pulse_length	Mandatory	float	Time length of source pulse [s], where 95% of the energy is included	e.g.: 0.01
serial_number	Optional	string	Serial number of sub-bottom profiler	
source_depth	Mandatory	float	Depth of sub-bottom profiler below the water surface [m]	e.g.: 1.5
source_level_Lpeak	Optional	integer	Zero-to-peak level [dB re 1 µPa] at 1m distance	e.g.: 190
source_level_SEL	Optional	integer	Sound-exposure level [dB re 1 µPa^2s] at 1m distance	e.g.: 175
/transects	Mandatory	group		

ICES_SubRectangles_Airgun	Mandatory	string(number of entries x 3)	Matrix, on which days in which ICES rectangles airgun sound events took place and in which category their zero-to-peak level falls. Days with several ICES rectangles are listed several times. Times are to be given in UTC DateTime in ISO 8601 format; YYYY-MM-DD. In case of varying zero-to-peak levels, use the maximum level for each line.	e.g.: [2021-09-17, 38F7, low; 2021-09-18, 38F7, very_low; 2021-09-18, 37F7, medium; 2021-09-19, 37F6, high; 2021-09-20, 36F6, NA ...]
ICES_SubRectangles_Other	Mandatory	string(number of entries x 3)	Matrix, on which days in which ICES rectangles boomer, sparker or sub-bottom profiler sound events took place and in which category their zero-to-peak level falls. Days with several ICES rectangles are listed several times. Times are to be given in UTC DateTime in ISO 8601 format; YYYY-MM-DD. In case of varying zero-to-peak levels, use the maximum level for each line.	e.g.: [2021-09-17, 38F7, low; 2021-09-18, 38F7, very_low; 2021-09-18, 37F7, medium; 2021-09-19, 37F6, high; 2021-09-20, 36F6, NA ...]
transect_count	Mandatory	integer	Number of transects	e.g.: 7
/transect_1	Mandatory	group	Further transects are indicated by increasing the number	e.g: [transect_2]
comment	Optional	string		
datetime	Mandatory	float(location_count x 1)	DateTimes of location coordinates, posixtimestamp in UTC	e.g.: [1630488180; 1630489080; 1630489980; ...]
location	Mandatory	float(location_count x 2)	Coordinates every 15 min, WGS84 Decimal Degrees, 5 decimal places, [lat, lon]	e.g: [55.62571, 4.09852; 55.62252, 4.09931; ...]
location_count	Mandatory	integer	Number of coordinates in location	e.g.: 54
name	Optional	string	Name of transect	
/sound_sources	Mandatory	group		
airguns	Optional	float(number of airguns x 1)	airgun labels of simultaneously operating airguns in this transect	e.g: [airgun_1; airgun_2a; airgun_5]
boomer	Optional	float(number of boomers x 1)	boomer labels of simultaneously operating boomers in this transect	e.g: [boomer_2; boomer_3a]
sparker	Optional	float(number of sparkers x 1)	sparker labels of simultaneously operating sparkers in this transect	e.g: [sparker_1b; sparker_2a]
sub_bottom_profiler	Optional	float(number of sub_bottom_profilers x 1)	sub-bottom profiler labels of simultaneously operating sub-buttom profilers in this transect	e.g: [sbp_3b]

HDF5-Beispiel

Dies ist ein Beispiel der HDF5-Struktur einer seismischen Ausfahrt. Metadatenfelder wurden für eine bessere Übersicht durch Punkte ersetzt. Bei dieser beispielhaften Ausfahrt waren zwei Airguns, ein Boomer und ein Sparker aktiv, wobei der Boomer in zwei unterschiedlichen Konfigurationen genutzt wurde. In Fahrtabschnitt 1 war die Arigun 1 aktiv, in Fahrtabschnitt 2 wurden beide Airguns genutzt. In Fahrtabschnitt 3 wurden gleichzeitig der Boomer in Konfiguration a und der Sparker genutzt. Im letzten Fahrtabschnitt waren wieder Boomer und Sparker im Einsatz, wobei für den Boomer die Konfiguration b genutzt wurde.

Field	Status	Data type	Field definition	Value
/				
...	...	...	...	...
/airguns	Optional	group		
airgun_count	Mandatory	integer	Number of airguns and airgun configurations	2
/airgun_1	Optional	group	Different airgun configurations are indicated by adding a letter	
...	...	...	...	...
/airgun_2	Optional	group	Different airgun configurations are indicated by adding a letter	
...	...	...	...	...
/boomer	Optional	group		
boomer_count	Mandatory	integer	Number of boomers and boomer configurations	2
/boomer_1a	Optional	group	Different boomer configurations are indicated by adding a letter	
...	...	...	...	...
/boomer_1b	Optional	group	Different boomer configurations are indicated by adding a letter	
...	...	...	...	...
/sparker	Optional	group		
sparker_count	Mandatory	integer	Number of sparkers and sparker configurations	1
/sparker_1	Optional	group	Different sparker configurations are indicated by adding a letter	
...	...	...	...	...
/transects	Mandatory	group		
...	...	...	...	...
transect_count	Mandatory	integer	Number of transects	4
/transect_1	Mandatory	group	Further transects are indicated by increasing the number	
...	...	...	...	...
/sound_sources	Mandatory	group		
airguns	Optional	float(1 x 1)	airgun labels of simultaneously operating airguns in this transect	[airgun_1]
/transect_2	Mandatory	group	Further transects are indicated by increasing the number	
...	...	...	...	...
/sound_sources	Mandatory	group		
airguns	Optional	float(2 x 1)	airgun labels of simultaneously operating airguns in this transect	[airgun_1; airgun_2]
/transect_3	Mandatory	group	Further transects are indicated by increasing the number	
...	...	...	...	...
/sound_sources	Mandatory	group		
boomer	Optional	float(1 x 1)	boomer labels of simultaneously operating boomers in this transect	[boomer_1a]
sparker	Optional	float(1 x 1)	sparker labels of simultaneously operating sparkers in this transect	[sparker_1]
/transect_4	Mandatory	group	Further transects are indicated by increasing the number	
...	...	...	...	...
/sound_sources	Mandatory	group		
boomer	Optional	float(1 x 1)	boomer labels of simultaneously operating boomers in this transect	[boomer_1b]
sparker	Optional	float(1 x 1)	sparker labels of simultaneously operating sparkers in this transect	[sparker_1]